



| EDUCATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                  |                         |      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-------------------------|------|
| Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Degree                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Institution                                                                      | % / CGPA                | Year |
| Dual Degree<br>(B.Tech + M.Tech)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Engineering Design,<br><i>Specialization - Robotics</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Indian Institute of Technology Madras, Chennai                                   | <a href="#">8.14/10</a> | 2024 |
| Class XII                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | CBSE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Maharishi Vidya Mandir, Chennai                                                  | <a href="#">95.4%</a>   | 2019 |
| Class X                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | CBSE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Maharishi Vidya Mandir, Chennai                                                  | <a href="#">10/10</a>   | 2017 |
| <ul style="list-style-type: none"><li>• Awarded (among <b>30</b> out of 800+ students) the IIT Madras <a href="#">Young Research Fellowship</a> based on remarkable research potential. 2021</li><li>• Secured <a href="#">97.6457 Percentile in JEE Advanced</a> and <a href="#">99.1046 Percentile in JEE Mains</a> out of over <b>1.5 Million</b> aspirants across India. 2019</li><li>• Secured an All India Rank <a href="#">of 52 in UCEED</a> (Undergraduate Common Entrance Examination for Design) conducted by <b>IIT Bombay</b>. 2019</li></ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                  |                         |      |
| RELEVANT COURSES AND SKILLS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                  |                         |      |
| Courses                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Robotics:</b> Mechanics and Control of Serial Robots (ED6007), Introduction to Robotics (ID6040), Introduction to Field and Service robotics (ED5315), Robotics Laboratory (ID6100)<br><b>Controls:</b> Control Systems (ED2040), Control of Automotive Systems (ED5330), Mechatronics System Design (ED5080), Analog and Digital Electronics (ED2130), Fundamentals of Automotive Systems (ED2160),<br><b>AI/SDE:</b> Fundamentals of Deep Learning (CS6910), Introduction to Computation and Visualization (ED1021), Transportation Network Analysis (CE5290), Data Structures and Algorithms (CS2910),<br><b>Math:</b> Series and Matrices (1102), Differential Equations (MA2020),<br><b>Product:</b> Entrepreneurship for Scientists and Engineers (ID5031), Functional and Conceptual Design (ED1011), Human Factors in Design (ED3010), Electromagnetic Compatibility for Product Design (ED5052), Digital Signal Processing for Engineering Design (ED5017), Fundamentals of Operational Research (MS3510),<br><b>CAD/Mech:</b> Geometric Modelling and CAD (ED2090), Design of Mechanical Systems 1 (ED2011), Design of Mechanical Systems 2 (ED4040), Energy Storage Devices and Systems (ED5011), |                                                                                  |                         |      |
| MOOCs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <a href="#">Algorithmic Toolbox*</a> (UC San Diego), <a href="#">Intro to tensorflow</a> , <a href="#">Reinforcement Learning</a> (Alberta), <a href="#">Game Dev.(Michigan)</a> , Economics of Banking and Finance (NPTEL – In grade card)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                  |                         |      |
| Skills                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Languages:</b> Python, C, C++, C#                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Tools:</b> VS Code, MATLAB, LaTeX, MS-Office, Unity3D, Fusion360, Linux       |                         |      |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Frameworks:</b> Tensorflow, Pytorch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Libraries:</b> Pandas, NumPy, Scikit Learn, Matplotlib, OpenCV, PyQt5, OpenGL |                         |      |
| RESEARCH AND PROFESSIONAL EXPERIENCE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                  |                         |      |
| <a href="#">DDP</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Yet to decided on topic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                  |                         |      |
| <a href="#">Systemantics</a><br>Robotics Design Intern                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                  |                         |      |
| <a href="#">Senai Lab</a><br>Computer Vision Intern<br>Jun 2022 – Present<br><a href="#">Proof 2</a> , <a href="#">Proof 3</a>                                                                                                                                                                                                                                                                                                                                                                                                                             | Project aims to develop a self-balancing & autonomous bicycle under the guidance of Prof. Raghunathan Rangaswamy <ul style="list-style-type: none"><li>• A</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                  |                         |      |
| <a href="#">Yaskawa Pvt. Ltd</a><br>Robotics Intern<br>Jun 2022 – Present<br><a href="#">Proof 2</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                  |                         |      |
| <a href="#">Young Research Fellow</a><br>Undergraduate Researcher<br>Jun 2021 – Present<br><a href="#">Proof 2</a>                                                                                                                                                                                                                                                                                                                                                                                                                                         | Research on Path Planning Algorithms for Unmanned Aerial Vehicles under the guidance of Prof. Satadal Ghosh. <ul style="list-style-type: none"><li>• Analyze</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                  |                         |      |
| <a href="#">Blurgs</a><br>Computer vision Intern<br>Jun 2021 – Aug 2021                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Employs real-time vision processing techniques to assist, automate, and augment the capabilities of drones in action. <ul style="list-style-type: none"><li>• Analyzed and tested <b>7+</b> State of the art <b>Multiple Object Tracking</b> networks for the startup's <b>first product</b>.</li><li>• Trained and tested various MOT networks like <b>Siam-MOT</b> and <b>FairMOT</b> for Surveillance with Drones and CCTV.</li><li>• Used <b>FairMOT</b> to track and get the trail of multiple moving objects and <b>Re-identify</b> objects in real-time.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                  |                         |      |
| <b>Co-curricular Affairs Secretary</b><br><b>Avishkar Hyperloop</b><br>Power Systems Engineer<br>Oct 2020 – Jun 2021                                                                                                                                                                                                                                                                                                                                                                                                                                       | The leading Asian student competition team with the vision to make Hyperloop transportation a reality in our country <ul style="list-style-type: none"><li>• <b>Winner</b> of the <b>most scalable design</b> award by Zeleros<sup>1</sup> and among the <b>top 5</b> for mechanical and propulsion.</li><li>• Designed a batter pack with <b>high discharge rate</b> for higher speeds while ensuring safe power supply to the pod</li><li>• Worked on a <b>custom-made Battery Management System</b> to maintain voltage, temperature and ensure safety.</li><li>• Extended battery pack life by <b>30%</b> by reducing peak operational temperature by <b>10°C</b> using an <b>Al<sup>4</sup></b> heat sink.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                  |                         |      |
| <a href="#">Imagine Pvt. Ltd.</a><br>Software Dev. Intern<br>Apr 2020 – Aug 2020                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Enables collaboration across different locations using VR/AR for purposes such as industrial design, support and sales. <ul style="list-style-type: none"><li>• Developed a <b>Unity3D</b> plugin to <b>upload any 3D model</b> into a VR<sup>2</sup> space using Autodesk <b>Forge APIs</b><sup>3</sup> in <b>real-time</b>.</li><li>• Built a basic <b>Heroku</b> app to upload and view any 3D model with <b>60+</b> file formats in the web browser.</li><li>• Achieved <b>100%</b> reduction in staff-hours required for this specific task by automating the entire pipeline.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                  |                         |      |
| PROJECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                  |                         |      |

|                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>VR – RL Shooter<br/>CVI<sup>8</sup> Club<br/>Co-curricular Affairs<br/>Secretary</b><br><i>(May 2020 – Jul 2020)</i>       | <ul style="list-style-type: none"> <li>• <b>Built and trained</b> a <b>Multi agent Reinforcement Learning</b> environment where agents play a shooting game.</li> <li>• The environment was made on <b>Unity3D</b> with <b>C#</b>, intractable blocks were included to see exploratory tactics.</li> <li>• Environment is setup to be ported as a <b>Virtual Reality</b> environment where the user can also play the game.</li> </ul>                      |
| <b>Path Finder  PyQt5<sup>6</sup></b><br><i>ED3150 (Apr 2021)</i>                                                             | <ul style="list-style-type: none"> <li>• Developed a 2D Path finder from scratch in Python using SOTA<sup>4</sup> <b>A-Star</b> and <b>Dijkstra</b> Algorithms and PyQt5.</li> <li>• Designed an <b>interactive UI<sup>5</sup></b> for visualization using <b>QT Designer<sup>6</sup></b>; Documented the project for future use.</li> </ul>                                                                                                                |
| <b>Control Systems</b><br><i>(ED2040)</i>                                                                                     | <ul style="list-style-type: none"> <li>•</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Brake Control (FAS)</b><br><i>ED5160</i>                                                                                   | <ul style="list-style-type: none"> <li>•</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>IFSR</b><br><i>ED5315</i>                                                                                                  | <ul style="list-style-type: none"> <li>•</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>SOC Prediction</b><br><i>ED5011</i>                                                                                        | <ul style="list-style-type: none"> <li>•</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Control of Automotive<br/>System</b><br><i>ED5330</i>                                                                      | <ul style="list-style-type: none"> <li>•</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>CS6910</b><br><i>CS6910 (Apr 2021)</i>                                                                                     | <ul style="list-style-type: none"> <li>•</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>CFD<sup>7</sup> Project (ED4040)</b><br><i>(Mar 2021 - Apr 2021)</i>                                                       | <ul style="list-style-type: none"> <li>• Designed an <b>aero foil</b> for the Front wing of a Formula Student car for maximum <b>Downforce/Drag</b> ratio.</li> <li>• Improved Force/Drag ratio by <b>33%</b> by varying position and angle of the aero foil relative to ground and car.</li> <li>• Performed <b>CFD<sup>7</sup> analysis</b> to determine maximum possible ratio and the optimal setup for better efficiency.</li> </ul>                   |
| <b>Human Powered<br/>Segway (ED4060)</b><br><i>(Apr 2021 – Jun 2021)</i>                                                      | <ul style="list-style-type: none"> <li>• Ideated a <b>novel</b> design of a fully mechanical version of a Segway under the guidance of Prof. Jayaganthan</li> <li>• Improved efficiency by <b>67%</b> by replacing conventional circular pedaling system with rectilinear pedaling.</li> <li>• Varied the number of linkages and lengths using MATLAB to optimize the design and improve its potential.</li> </ul>                                          |
| <b>Gesture Robot</b><br><i>ED5080 Project<br/>(July 2021- Nov 2021)</i>                                                       | <ul style="list-style-type: none"> <li>• Designed a robotic arm with 2 DOF 8 for identifying and lifting objects using electromagnets.</li> <li>• Implemented proximity sensors for detection and used LED light reflection for sensing color.</li> <li>• Programmed using Arduino mega and added manual and automatic calibration features.</li> </ul>                                                                                                     |
| <b>Product Design</b><br><i>ED1011</i>                                                                                        | <ul style="list-style-type: none"> <li>• Analysed various products and deduced 50+ design changes for maximum customer satisfaction and usability.</li> <li>• Ideated all the necessary parameters and metrics to approach the new design plan for a given product.</li> <li>• Used different product design tools such as Flow method, TRIZ, Product architecture, House of quality etc</li> </ul>                                                         |
| <b>POSITION OF RESPONSIBILITIES</b>                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Academic Affairs<br/>Secretary<br/>Placement Core</b><br><i>(May 2021 - Present)</i>                                       | <ul style="list-style-type: none"> <li>• Member n</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Academic Affairs<br/>Secretary<br/>Industrial and Public<br/>Relations, Coordinator</b><br><i>(May 2021 - Present)</i>     | <ul style="list-style-type: none"> <li>• Member of team managing <b>200+ non-engineering profiles</b> in sectors such as <b>consulting, finance</b> and <b>analytics</b></li> <li>• Discovered and contacted <b>20+</b> potential recruiters based on skill-set requirements and student requests</li> <li>• Responsible for coordinating logistics of <b>100+ firms</b> and <b>1000+ final year students</b> in the upcoming season</li> </ul>             |
| <b><a href="#">ShARE, Junior<br/>Consultant</a></b><br><i>(May 2021 – Present)</i>                                            | <ul style="list-style-type: none"> <li>• Selected out of <b>200</b> applicants after multiple rounds of <b>Global Interview</b> comprising of consulting case analysis</li> <li>• Working in '<b>Mobility</b>' Module, undergoing training on focused application of problem-solving in consulting</li> <li>• Part of a <b>global team of 700 students</b> across <b>15 nations</b>, enabling firms to transition into '<b>Do Well, Do Good</b>'</li> </ul> |
| <b>Academic Affairs<br/>Secretary<br/>Internship Team,<br/>Coordinator</b><br><i>(May 2020- Apr 2021)</i>                     | <ul style="list-style-type: none"> <li>• Organized <b>first-ever virtual Internship Day-1</b> achieving <b>152</b> offers from <b>20</b> companies across sectors</li> <li>• Part of a team of 25 <b>liaising with companies</b> across multiple sectors for Campus Internships on a rolling basis</li> <li>• Acted as a single Point of Contact for <b>20+</b> companies, interacting with <b>600+</b> students across departments.</li> </ul>             |
| <b>Co-curricular Affairs<br/>Secretary<br/>CVI<sup>8</sup> Club, Coordinator</b><br><i>(May 2020- Apr 2021)</i>               | <ul style="list-style-type: none"> <li>• Responsible for the club activities like conducting peer-to-peer sessions, mini-projects, events and projects.</li> <li>• Conducted student run sessions where basics of OpenCV, Machine Learning &amp; Deep Learning were taught.</li> </ul>                                                                                                                                                                      |
| <b>Co-curricular Affairs<br/>Secretary<br/>Coding and Logic<br/>Events, Coordinator</b><br><i>(May 2020- Apr 2021)</i>        | <ul style="list-style-type: none"> <li>• Handled a total footfall of <b>300+ teams</b> and crafted a total of <b>50+ logical questions</b> for events in Shaastra'21<sup>8</sup>.</li> <li>• Assisted the team in the successful completion of Hackathons conducted by <b>AstraZeneca</b> and <b>Atlassian</b></li> <li>• <b>Ideated</b> and executed the conduction of <b>2 brand new</b> events, Mystery Rooms and E-Contest</li> </ul>                   |
| <b>Co-curricular Affairs<br/>Secretary<br/>Associate Manager,<br/>Startup Services, E-Cell</b><br><i>(Nov 2019- Apr 2021)</i> | <ul style="list-style-type: none"> <li>• Organized Summer Internfair 2020, handling logistics, interviews and congregating around <b>40+</b> Startups.</li> <li>• Organized <b>Elevate</b>, a fund-raising competition for startups, bringing in <b>150+</b> startups from all over the country</li> <li>• Managed the operational requirements of board members of unicorns throughout the Annual <b>E-Summit<sup>10</sup></b></li> </ul>                  |
| <b>EXTRA CURRICULAR ACTIVITIES</b>                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|                                                    |                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Drawing</b>                                     | <ul style="list-style-type: none"> <li>• Won <b>3</b> state-level competitions and 5+ District level competitions in drawing (Oil Pastels and Pencils)</li> </ul>                                                                                                                                       |
| <b><a href="#">Teaching Assistant (ED5315)</a></b> | <ul style="list-style-type: none"> <li>• Responsible for giving, evaluating and</li> </ul>                                                                                                                                                                                                              |
| <b><a href="#">Teaching Assistant (GN1101)</a></b> |                                                                                                                                                                                                                                                                                                         |
| <b>Teaching assistant (GN6106 (HHS))</b>           | Happiness Habits and success (course name proof in grade card)                                                                                                                                                                                                                                          |
| <b>Game Dev. / Gaming</b>                          | <ul style="list-style-type: none"> <li>• Love to try out various tools and features in Unity3D; Reached a rank of <b>Silver 2</b> in <b>Valorant</b> by Riot Games.</li> <li>• Solely developed a <b>complete 3D Game</b> in Unity3D with Obstacles, Enemies, Points, UI and Visual Effects.</li> </ul> |
| <b>Sports</b>                                      | <ul style="list-style-type: none"> <li>• Among the <b>20 out of 100</b> students selected for freshman Volleyball training (NSO);</li> </ul>                                                                                                                                                            |
| <b><a href="#">Fusion360 API</a></b>               | <ul style="list-style-type: none"> <li>• Developed a <b>python script</b> to automatically create models in <b>Fusion360</b> just by specifying dimensions.</li> </ul>                                                                                                                                  |

\* 8.51 : As of 20 Feb '22; [1] Zeleros – A Spanish company that develops Hyperloop; [2] Virtual Reality; [3] Application Programming Interface; [4] State of the Art ; [5] User Interface; [6] Tool for designing and building graphical user interfaces ; [7] Computational Fluid Dynamics; [8] Computer Vision and Intelligence Club; [9] ISO Certified Annual Technical Festival of IIT Madras; [10] ISO Certified Annual Entrepreneurship Summit; \*- ongoing courses